

Session 12 — Deploy and read your code

In class (90 min). Type along with the instructor. The project is **not graded**. The final exam is a closed-book MCQ: you must be able to **explain every line** you type.

Goal. Freeze features, walk through Streamlit Community Cloud, practise exam-style reading.

You leave with. You know the Cloud settings and why bundled data/ matters. There are **no student demos**.

0.1 Feature freeze

Do **not** add tabs, heatmaps, or extra tickers today. Commit a clean tree:

```
git status
git add -A
git commit -m "session 12: freeze"
git push
```

Your GitHub repo must be visible to Streamlit (usually **public**, or Cloud connected to your GitHub account).

0.2 How Cloud differs from your laptop

On your laptop you often run:

```
uv run streamlit run dashboard.py
```

from the folder that **contains** `dashboard.py`.

Streamlit Cloud clones the **GitHub repository root**.

- If the repo **is** the dashboard (template): **Main file path** is `dashboard.py`.
- If the dashboard lives in a subfolder `dashboard/`: **Main file path** is `dashboard/dashboard.py`.

Python version: **3.12** (not 3.14). Dependencies: committed `uv.lock` and/or `requirements.txt` in the folder Cloud inspects (follow the instructor).

`DATA_DIR` uses `Path(__file__)` in `paths.py`, so bundled CSVs still resolve on Cloud. **Never** write into `data/` at runtime.

Yahoo Finance is often **blocked** on Cloud. The app must still render from CSVs. That is a feature, not a failure.

0.3 Walkthrough (follow the shared screen)

1. Open share.streamlit.io and sign in with GitHub.
2. **Create app**.
3. Repository: your fork (or the volunteer repo the instructor uses).
4. Branch: `main` (unless you were told otherwise).
5. **Main file path**: as above.
6. Python: **3.12**.
7. Deploy. Watch the **build log**. Red text about Yahoo is expected; a crash on **import** is not.

Note. No secrets. FRED uses the public `pandas_datareader` feed.

If Cloud is down, rehearse locally from the **repository root** (instructor course repo layout):

```
uv run --project dashboard streamlit run dashboard/dashboard.py
```

0.4 Troubleshooting

Symptom	Check
Module not found	Is <code>uv.lock</code> committed? Did Cloud pick the dashboard project?
Wrong Python	App settings → 3.12

Theme missing	<code>.streamlit/config.toml</code> next to the app (or repo root — follow the instructor)
Page not found	<code>dashboard.py</code> builds view paths from <code>Path(__file__).parent</code>
Empty charts	Bundled data/ present? Live Yahoo/FRED failed — look for your <code>st.info</code> banner

0.5 Exam-shaped reading (in class)

The final is **closed book**, snippet + true/false. Open **your** `views/common.py` and `views/equities.py`. With a neighbour, answer out loud:

1. After `@st.cache_data` on `_fetch_live_prices`, does the second identical call hit Yahoo again within the TTL?
2. Why is Yahoo's end one day after the last date we want on the chart?
3. What does `preprocess` do if a column has a hole, then a value? (`.ffill()` then `.dropna()`)
4. Why fetch `end_date - 365 days` even when the chart starts yesterday?
5. Why must Bonds use `bonds_start_date` instead of `start_date`?

If you built FX: what is the difference between `usd_quote` and `usd_base`?

You will not get a quiz paper today. The **habit** is the point: read the line, predict, then look.

0.6 Optional: tests

If `tests/test_loaders.py` exists:

```
uv run pytest
```

A passing test is nice. It is not a grade.

0.7 After class

The project stays ungraded. Revise **your** functions for the final MCQ. AI is allowed while you work **if you can explain every line**; the exam has no AI.